

Laboratory Management System Database Implementation Report

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1 Team Roles

- **Abdelhalim (Lead Developer):** Translated conceptual model structures into MySQL DDL scripts, handled data types, defined table constraints, and eliminated SQL syntax errors during implementation.
- **Hamza & Reda (Co-Requirement Analysts & Conceptual Modelers):** Analyzed domain requirements, designed the initial conceptual ER model, mapped relationship hierarchies and aggregations, and ensured schema integrity.

2 Key Challenges & Solutions

1. **Self-Referential Supervision (Employee):** Represented 1-to-N recursive supervision by adding a nullable foreign key `supervisor_pid` referencing `Employee(pid)`. This allows top-level managers without causing foreign key cycles.
2. **Aggregation & Ternary Relationships:** Researched aggregation mapping rules via external sources to represent multi-entity interactions (e.g., consumable usage per reservation and lab stock). Mapped these interactions using associative tables with 3-attribute composite primary keys, such as `Consumes(consId, labId, resId)`.
3. **Identifier Syntax Conflicts:** Resolved SQL parser errors by wrapping table names and column identifiers in backticks (``...``) to prevent conflicts with reserved SQL keywords.

3 Justification of Conversion Choices

- **Class Hierarchies (Table-per-Type):** Subtyped entities like `Person` and `Employee` into dedicated subtype tables using shared primary keys (`pid`) as foreign keys. This keeps tables normalized and avoids large numbers of NULL attributes.
- **Selective Cascade Deletes:** Applied `ON DELETE CASCADE` specifically to dependent tracking tables (`Maintenance`, `CalibrationRecord`) so associated logs are cleaned up when an equipment unit is deleted. Kept parent entity references at default (`NO ACTION`).
- **Composite Keys for Associations:** Used multi-attribute composite primary keys in associative tables (`Advises`, `Stocks`, `Supplies`) to enforce uniqueness across multi-way relationships without generating unnecessary surrogate keys.

4 System Assumptions

- **Non-Null Composite Keys:** Every attribute forming a primary key or composite primary key tuple is non-null.
- **Stock Inventory Dependency:** A consumable usage entry (`Consumes`) assumes the item is tracked in active lab inventory (`Stocks`).

- **Unit-Level Reservation Tracking:** Equipment reservations (`Reservation`) map to physical equipment units (`EquipementUnit`) via unique serial numbers rather than generic equipment models.